

PM Project 4: Qualitative Strategy Brief

Why robust_growth_v2 is the leading candidate

Recommendation. Review robust_growth_v2 first. This brief gives no empirical performance claims. It states the design logic and the hypotheses that recipients should verify independently.

Objective

Maximize after-cost compound growth while avoiding ruin. The first position is free; later trades face costs, a per-security daily cap, and cap penalties. Existing positions drift before each new trade, so a large loss can increase actual gross while the cap slows deleveraging.

Hypotheses to verify

Signal 3 should be tested as a relatively short-horizon signal. Signals 1 and 2 should be tested for slower persistence. A linear blend should be compared with powers, squares, logs and products. A simple diversified core should be compared with inverse-volatility and covariance-estimated cores. None of these patterns is asserted here as a fact.

Exact selected construction

Cross-sectionally standardize each signal and lag it one day. Smooth signals 1/2/3 with half-lives 5/15/5 and mix them 30%/40%/30%. Combine a 52.5% equal-weight core with a 47.5% unit-gross signal book. Estimate downside risk over 60 days and set leverage to clip(0.0375 / downside risk, 2, 6), with a small causal common-signal-2 adjustment. Cap ideal positions at 1.5 per security.

Execution controls

Track post-cost drift internally. Apply a volatility deadband before trading, clip planned changes to 0.049, and cap submitted gross at 9.5. These controls aim to reduce cost, avoid routine cap penalties, and prevent targets from becoming detached from the book the evaluator actually holds.

Why the rule is linear

Signals can be negative, making ordinary logs unsuitable. Unsigned squares remove direction. Powers and products can magnify extreme scores and concentration. A linear blend is easier to audit and less sensitive to one unusual observation.

Finalist alternatives

linear_no_timing removes the least certain market-timing term. linear_lower_tilt reduces dependence on signal stability. linear_conservative lowers leverage and concentration. linear_fixed_5_5 is a simple fixed-leverage benchmark. equal_core_5_5 is the signal-free control. These are genuine alternative decisions, not cosmetic variants.

Main risks

Signal decay or reversal; regime dependence; underestimated tail dependence; false confidence from parameter selection on a short sample; and post-loss gross exposure that cannot be reduced quickly because of the trade cap. No backtest or bootstrap proves zero bust probability.

Independent review checklist

Recompute signal information by forecast horizon and signal persistence. Use rolling and blocked validation. Destroy and reverse signal alpha. Stress persistent volatility, reduced drift and near-ruin losses. Inspect actual gross after drift. Verify truncation causality and the exact cost/cap evaluator. Prefer a reproducible counterexample over a general preference for lower volatility, because the project scores compound growth.

The accompanying Python files are authoritative. The editable LaTeX source is included beside this PDF.